

Muhammad Abdullah

Full-Stack Developer | Cybersecurity | Data Analytics

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PROFESSIONAL SUMMARY

Final-year BSc (Hons) Computer Science student at the University of West London (graduating 2026) with end-to-end experience shipping production SaaS, conducting mobile penetration testing, and delivering multi-environment data analytics. Built and deployed CareerCompass — a live React + Flask + PostgreSQL platform serving real users with NLP-driven career recommendations and Stripe billing. Completed a rigorous data analytics project across three environments (Python, R/SQL, MongoDB Atlas) analysing 1,250+ orders with actionable operational insights. Conducted OWASP MASVS-aligned Android penetration testing with full remediation documentation. Actively seeking graduate or junior roles in Software Engineering, Full-Stack Development, Data Engineering, or Cybersecurity across the UAE.

TECHNICAL SKILLS

Languages	Python · TypeScript · JavaScript · Java · Kotlin · SQL · R · HTML5 · CSS3
Frontend	React · TypeScript · Responsive UI · Figma · Accessibility (WCAG)
Backend & APIs	Flask · Node.js · REST API Design · Stripe (Checkout, Webhooks, Billing Portal) · Authentication · RBAC
Databases	PostgreSQL (Row-Level Security) · Supabase · SQLite · MongoDB Atlas · DBI/RSQLite
Data & Analytics	pandas · NumPy · scikit-learn · spaCy · BART-MNLI · ggplot2 · seaborn · matplotlib · Jupyter · Google Colab
Cybersecurity	OWASP MASVS · Mobile Top 10 · Penetration Testing · Kali Linux · ADB · Android Keystore · AES-256-GCM · NIST SP 800-57 · Cryptography · Secure Coding
Software Eng.	OOP · SOLID · Design Patterns (Observer, Repository) · UML · OCL · JUnit 4 · TDD · Agile/Scrum · Risk Management
DevOps & Tools	Git · GitHub · CI Workflows · Docker · Vercel · Render · Android Studio · VS Code · Cisco Packet Tracer

PROJECTS & EXPERIENCE

Full-Stack Developer · **CareerCompass — Final-Year Dissertation (Live SaaS)**
careercompass-iota.vercel.app

2024 – 2026

React · TypeScript · Flask (Python) · PostgreSQL · Supabase · Stripe · spaCy · BART-MNLI · Vercel · Render · Git

- Architected and shipped a production SaaS platform end-to-end, delivering personalised, explainable career guidance to real users via a React/TypeScript SPA and Flask REST API backend.
- Engineered an explainable recommendation engine combining Big Five (OCEAN) personality scoring, resume skill extraction, and interest mapping to produce ranked career match scores (0–100) with human-readable rationales.
- Built ATS-style resume feedback using spaCy NLP and BART-MNLI zero-shot classification; integrated live job search, application tracking, and multi-tenant data isolation via PostgreSQL Row-Level Security.
- Implemented Stripe subscription billing across Free / Pro / Enterprise tiers with Checkout, signed webhook verification, and a customer billing portal — achieving fully automated upgrade, downgrade, and cancellation flows.
- Deployed continuously to Vercel (frontend) and Render (backend); validated with unit, integration, and end-to-end functional tests against the live production environment.

Data Analyst · NorthStar Urban Mobility & Logistics — Academic Project

Jan – May 2026

Python 3.12 · R · SQL · SQLite · MongoDB Atlas · pandas · NumPy · ggplot2 · seaborn · Google Colab

- Analysed 9 operational datasets (1,250 orders, 950 deliveries) across three environments — R/SQL, Python, and MongoDB Atlas — for a simulated multi-zone urban logistics company.
- Authored 5 production-style SQL queries in R (DBI/RSQLite) covering service revenue, delivery cost, problem-rate by service type, hub performance, and booking-channel failure rates.
- Surfaced actionable operational insights: Hub H05 worst performer (3.30 hr avg delay, 3.66/5 rating); Central zone highest delay rate (35.29%); Business orders highest problem rate (32.12%).
- Cleaned and merged 9 raw datasets in pandas into a 1,250-row, 51-column master dataset; engineered delivery-duration and delay-flag features for downstream modelling.
- Designed a MongoDB Atlas document schema with nested records; executed 5 CRUD operations and 6 aggregation pipelines; recommended a hybrid relational + document architecture.

Cybersecurity Analyst · Android Mobile Security Assessment — Academic Project

2025 – 2026

Kotlin · Kali Linux · ADB · Android x86 · Jetpack Security · OWASP MASVS · AES-256-GCM · Android Keystore

- Built a custom Android application to reproduce OWASP M9 (Insecure Data Storage) in a controlled Kali Linux + Android x86 lab environment.
- Executed black-box penetration testing via ADB shell to extract plain-text credentials from SharedPreferences XML without root; confirmed data leakage through logcat analysis.
- Remediated the vulnerability using EncryptedSharedPreferences backed by the Android Keystore with AES-256-GCM; re-ran the attack to verify ciphertext was unreadable.
- Documented findings against OWASP MASVS and produced an 8-step security hardening guide covering encryption, log sanitisation, and backup restriction.

Cryptography Researcher · Extended Playfair Cipher — Academic Project

2025 – 2026

Cipher Design · Cryptanalysis · NIST SP 800-57 · OWASP

- Designed a 9×9 (81-cell) extended Playfair matrix supporting uppercase, lowercase, digits, and 19 symbols — resolving case loss, I/J collision, and filler ambiguity in the classic 5×5 design.
- Traced full encryption and decryption across four digraph pairs with annotated matrix diagrams, demonstrating 100% lossless recovery.
- Evaluated 5 cryptographic weaknesses against NIST SP 800-57 and OWASP guidance; benchmarked against bcrypt and Argon2id password hashing standards.

Software Engineer · Enterprise Knowledge Management System — Academic Project

2025 – 2026

Java · UML · OCL · Observer & Repository Patterns · JUnit 4 · OOP · SOLID

- Produced full UML design (use-case, class, sequence, component diagrams) for an enterprise KMS serving 68 global offices.
- Implemented in Java using OCL constraints, Observer and Repository patterns, and a Service Layer architecture aligned to SOLID principles.
- Delivered a 14-step JUnit 4 integration suite with OCL constraint validation, plus a complete software-maintenance analysis (corrective, adaptive, perfective, preventive).

Frontend Developer · Rivo Vogue eCommerce Website — Freelance

2023 – 2024

rivo-vogue1.odoo.com

Odoo CMS · HTML · CSS · Responsive Web Design

- Built and styled a live men's fashion eCommerce storefront on Odoo CMS, optimising responsive layout, product display hierarchy, and frontend performance.

UI/UX Designer · Smart Home Automation App — Academic Project

2023 – 2024

Figma · Wireframing · Interactive Prototyping · Accessibility · User-Centered Design

- Designed a fully interactive mobile prototype featuring accessible smart home controls, intuitive navigation flows, and a user-friendly dashboard layout.

Network Engineer · Global Network Infrastructure Simulation — Academic Project

2023 – 2024

Cisco Packet Tracer · IP Addressing · Subnetting · Routing Protocols

- Simulated a multi-country enterprise network with complete IP addressing, subnetting design, and routing protocol configuration for international connectivity.

EDUCATION

BSc (Hons) Computer Science · University of West London

2023 – 2026 (Final Year)

- Modules: Software Engineering · Cybersecurity · Databases & Analytics · OOP · Human-Computer Interaction · Networking.
- Dissertation: CareerCompass — production-deployed career guidance platform with NLP, explainable AI, and Stripe billing.

FSc Pre-Engineering · Punjab Group of Colleges, Lahore

2020 – 2022

Matriculation (IGCSE-equivalent) · Al Amaal English High School, Sharjah

2009 – 2020

LANGUAGES & INTERESTS

Languages: English (Fluent) · Urdu (Fluent)

Interests: Full-Stack Development · Cybersecurity · Data Engineering · Mobile Security · Open Source